9.6.2.1 Internal Memory

The memories which are kept inside the CPU (or) on mother board are known as Internal Memory. There are five types, such as ROM, RAM, PROM, EPROM and EEPROM. All are nothing but IC's. Let us see the purpose and usage of each memory device in following sections.

ROM →Read Only Memory

From the above expansion itself, we could understand that the content of this IC can be read only. The function and working of this IC can be easily explained through the following example.

Let we take a calculator:

Switch on the calculator.

What you will get on the screen?

'0'

But you didn't press zero. Then, how the zero comes?

Whether it came automatically?

No, someone is making it to display.

Just like the same, whenever we are switching on the computer, the computer has to do something on its own to get ready.

We know basically the computer cannot do anything on its own.

Then, it is doing something means, the necessary command is delivered from a particular memory. That particular memory is called as ROM.

This is an IC (Integrated Circuit), whenever we are switching ON the computer, the pointer will enter into this memory, read the content and do accordingly.Initially, it will do a self-test, which is called as POST(Power On Self-Test). After performing POST, it will hand over the charge to us. Then only, we can give anything to the computer.



POST - Who written and stored these necessary testing commands inside the ROM?

- It is written and stored by the manufacturer.

After storing the necessary instruction, the IC will be sealed. Then, the content cannot be altered, deleted and changed. But, can only be read. Because of this it is called as Read Only Memory or permanent memory. Without this ROM, no calculator or computer can get ready.

RAM →Random Access Memory

Random means without any proper order. Let we see the purpose of this memory. Let we again take calculator for example.

Assume that you want to perform the addition of 5, 10 in the calculator. What we will do?

Step1 : Switch ON the calculator.
:You can get '0' on the screen.

Step2 :Now press'5'.
:5 will be displayed on the screen.

Step3 : Next press '+'.

Step4 : For '10' you have to press'1'.

Now what will happen to that '5' it is not on the screen, then where it would be. Certainly, these numbers have gone to a memory and that particular memory is RAM. Since we cannot find the location

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